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Machine learning for predicting elections in Latin America based on social media engagement and polls

Kellyton Brito[a,*], Paulo Jorge Leitão Adeodato[b]

[a] Departamento de Computação, Universidade Federal Rural de Pernambuco, Recife, Brazil
[b] Centro de Informática, Universidade Federal de Pernambuco, Recife, Brazil

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ABSTRACT

The popularity of contemporary social media (SM) has impacted democratic practices, and the success of presidential campaigns is frequently attributed to SM performance. Within this new scenario, many methodological proposals that use SM data have been put forward for predicting election results. However, the most common approach, based on the volume and sentiment analysis of mentions on Twitter, has been frequently criticized and challenged. Thus, recent surveys have indicated new directions, such as the use of data from more than one SM platform, the adoption of nonlinear machine learning (ML) models, and the validation of methodologies and experiments in different elections. In this context, the present paper proposes SoMEN, the Social Media framework for Election Nowcasting, a framework composed of a process and an ML model for nowcasting election results based on SM performance as features and with offline polls as labeled data. It also defines SoMEN-DC, an execution strategy for SoMEN that enables continuous prediction during the campaign (DC). The proposed metrics and framework were applied to some of the main recent presidential elections in Latin America: Argentina (2019), Brazil (2018), Colombia (2018), and Mexico (2018). More than 65,000 posts were collected from the SM profiles of candidates on Facebook, Twitter, and Instagram with data from 195 presidential polls. Results have demonstrated that it was possible to achieve a high level of accuracy in predicting the final vote share of the candidates and to make daily predictions, providing competitive or better results than the traditional polls. The strategies put forward in this study have attempted to address several of the current challenges in this research area and have indicated a new manner of how to face the problems. They may also be directly used for predicting future elections in similar scenarios.

1. Introduction

Contemporary social media (SM) platforms are relatively new: Facebook was launched for public access in 2006, Twitter debuted in 2006, and Instagram emerged in 2010. This scenario is also constantly changing, e.g. TikTok was launched in 2016 and has already become very popular (Xu, Yan, & Zhang, 2019). Despite its novelty aspect, SM denotes a new communication paradigm, fundamentally altering the way that humans communicate (Kent & Li, 2020). Moreover, politicians are using the capabilities of SM in order to run permanent campaigns with neither geographic nor time constraints. Thus, the success of presidential campaigns is frequently attributed to being correlated with the SM performance of the candidates, such as the presidential campaigns in the U.S. of Barack Obama in 2008 and 2012 (Bimber, 2014; Cogburn & Espinoza-Vasquez, 2011) and Donald Trump in 2016 (Francia, 2018), and the Brexit campaign in the U.K. in 2016 (Hall, Tinati, & Jennings, 2018). Indeed, (Brito, Meira, & Adeodato, 2021) reported strong statistical correlations between the interactions received by the official profiles of candidates running for the Brazilian presidency in 2018 and the electoral results, and a follow-up study also found the same correlations in Argentina, Colombia, and Mexico (Brito & Adeodato, 2022). Within this context, researchers have tried to predict election results based on SM data.

Election polls are the de facto mechanisms for predicting political outcomes, and scientific polling dates back to 1936, after the first well-known polling crisis (Crossley, 1937). However, there are certain theoretical advantages of using SM data to gather information on citizens, with regard to its outreach, speed, and low cost. Face-to-face and telephone-based surveys are only able to reach a small fraction of the population, require time to be performed, and incur high costs. In

* Corresponding author.
E-mail addresses: kellyton.brito@ufrpe.br (K. Brito), pjla@cin.ufpe.br (P.J.L. Adeodato).

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practice, although in countries such as the U.S. traditional polls are conducted almost daily, in Latin American countries such polls are rare and are mostly concentrated towards the end of campaigns. However, in, 2020, SM platforms were being actively used by 51% of the world population (We are Social, and Hootsuite, 2020), and it is possible to gather and process data from these platforms for a fraction of the cost of traditional polls, and almost in real time.

In parallel, over the past two decades, the ongoing exponential increase in the availability of online data and low-cost computation, together with the development of new learning algorithms and theory, has led to the wide adoption of machine learning (ML) methods and techniques, which address the question of how to build computers that automatically improve through experience (Jordan & Mitchell, 2015). This has been used in a number of huge, complex data-intensive fields such as medicine, astronomy, biology, hydrology, finance, and economics (Ardabili, Mosavi, & Várkonyi-Kóczy, 2020; Qiu, Qihui, Ding, Yuhua, & Feng, 2016), since these techniques provide possible solutions to mine information hidden in data, which has impacted greatly on science and society (Rudin & Wagstaff, 2014).

This scenario has revealed SM as a natural candidate for being used as input data, and ML approaches, capable of mining information hidden in data, as a promising approach for performing predictions. However, this is not a traditional scenario of ML use. Despite the availability of SM data, its use presents a number of challenges, such as the possibility of being affected by volume manipulation together with the rapidly changing SM landscape. The political scenario also presents new challenges for ML, such as the lack of available historical data, since elections usually take place every four or five years, and the existence of just one actual labeled data, the final vote share, which is the aim of prediction.

1.1. Related work and the main challenges in predicting elections with social media data

Two studies may be considered seminal in this research topic. In 2010, Tumasjan et al. (Tumasjan, Sprenger, Sandner, & Welpe, 2010) claimed that "the mere number of tweets mentioning a political party can be considered a plausible reflection of the vote share and its predictive power even comes close to traditional election polls." In the same year and with an improved approach via a sentiment detection of tweets, O'Connor (O'Connor, Balasubramanyan, Routledge, & Smith, 2010) observed that "a relatively simple sentiment detector based on Twitter data replicates consumer confidence and presidential job approval polls."

Following these studies, much work has been published with the aim of predicting election results. Three very recent surveys have summarized these initiatives. Chauhan et al. (Chauhan, Sharma, & Sikka, 2021) surveyed 38 papers, Brito et al. (Brito, Silva Filho, & Adeodato, 2021) performed a systematic review including 83 studies, and Santos et al. (Santos, Bernardini, & Paes, 2021) analyzed 53 studies. With minor methodological differences, they all indicate that the most commonly used approach is still based on volume and sentiment analysis of Twitter posts, with the platform being used to poll the opinions of citizens. The usual process of this approach consists of (i) Twitter data collection by pre-selected keywords during a specific interval; (ii) data cleaning, by removing tweets not addressing elections, together with duplicates or retweets; (iii) sentiment analysis using lexicon-based approaches or ML approaches; (iv) prediction based on volume/sentiment counting analysis using a linear formula; and (v) performance evaluation. In most cases, prediction is a direct correlation between the percentage of posts mentioning a candidate and the candidate's vote share, and the performance evaluation varies from just indicating the winner to analyzing the vote share.

Table 1
Summary of Main Challenges in this research area (Brito et al., 2021).

Category	Challenge
Process	Lack of well-defined replicable processes
	Lack of generalization
	Lack of prediction capabilities during the campaign
Sampling	Social networks do not represent a good population sample
	Twitter is the most used but is not a good sample of social network
	Collected data on Twitter is not a good sample of Twitter data
Modeling	Arbitrary data collection choices
	High susceptibility to volume manipulation
	Difficulties in crossing data from multiple networks
	Lack of use of state-of-the-art machine learning
	Technical modeling weaknesses
Performance evaluation and Scientific rigor	Lack of statistical analysis of results
	Lack of meaningful comparison of results with related works
	Lack of discussion regarding bias and threats to validity

However, this approach engendered a number of criticisms just after it was launched (Gayo-Avello, 2011; Gayo-Avello, Metaxas, & Mustafaraj, 2011), and even the possibility of replication was contested. After replicating Tumasjan's seminal study, Jungherr (Jungherr, Jürgens, & Schoen, 2012) argued that "the results are contingent on arbitrary choices of the authors" and indicated that the mere inclusion of one more party or day of collection could greatly change the results. Current surveys have confirmed this low performance level: Brito et al. (Brito et al., 2021) reported that there was only a 55% success rate in the 64 studies that used the volume/sentiment approach. In a follow-up study (Brito, & dos Santos, and Paulo Jorge Leitao Adeodato., 2022), they directly argued that, due to several biases and challenges, using Twitter to poll the opinions of citizens may not be feasible at all. As a conclusion, they argued that researchers should attempt to use approaches based on multiple platforms, together with additional non-SM data, such as polls or economic indicators, as used in statistical forecasting (Lewis-Beck, 2005) models.

The recent surveys have also presented directions in four areas: processes, sampling, modeling, and performance evaluation and scientific rigor, as presented in Table 1. They indicate that there is a need for: well-defined, generalizable, and replicable processes, since most studies are applied in just one election; a robust methodology for data collection and analysis, due to the arbitrary choices of data collection; scientific rigor in order to analyze the results, since many studies claim success just by indicating the election winner or claiming that results were close to the electoral results; and the exploration of machine learning (ML) algorithms and models to tackle the peculiarities of this scenario, rather than only improving the sentiment analysis. Furthermore, the capability of predicting the electoral results during the campaign is also indicated as an open challenge, and the possibility of using platforms other than Twitter is highlighted.

Through an analysis of the related research, it is distinctly identified that this research topic presents many challenges. Indeed, different researchers applying the same approach in the same context may achieve opposite results (Jungherr et al., 2012; Tumasjan et al., 2010), and the same researchers applying the same approach in different contexts may

also achieve contradictory results (Anjaria & Guddeti, 2014; Goto & Goto, 2019). Lastly, predicting election results may be considered an activity of nowcasting, an estimation of the present or very near future, rather than a prediction or forecasting, since many voters only decide on their vote during the last few days before an election or even on the day of the election. Thus, this study focuses on presenting a new approach that considers these challenges, particularly those presented in Table 1.

1.2. The current study

The present work proposes SoMEN, the Social Media framework for Election Nowcasting, a framework composed of a process and model for nowcasting election results based on SM performance features and using ML approaches. It proposes a process with five well-defined steps, ranging from election understanding to prediction evaluation. It also uses a recently defined set of metrics based on Zajonc's exposition theory (Zajonc, 1968, 2001), and an ML model for predicting the final election results based on an ensemble of artificial neural networks (ANN) trained with SM metrics as features together with offline polls as labeled data. It also defines SoMEN-DC, an execution strategy for SoMEN that enables continuous prediction during the campaign (DC).

The proposed framework was applied to some of the main recent presidential elections in Latin America: Argentina (2019), Brazil (2018), Colombia (2018), and Mexico (2018). More than 65,000 posts were collected from the SM profiles of the candidates on Facebook, Twitter, and Instagram, starting from 300 days before election day, as well as data from 195 presidential polls conducted during the same period. We then statistically tested our results with pollster data from each election and compared our results with similar studies and the historical baseline from the polling industry (Jennings & Wlezien, 2018).

Results demonstrated that it was also possible to achieve a high level of accuracy in predicting the final vote share of the candidates, providing competitive or even better results than the traditional polls. Moreover, despite the difficulty involved in measuring the quality of predictions during the campaign, the results of daily predictions have been promising and competitive with polls.

This paper studies the central feature of many governments and democracies, the presidential elections. As contributions, we highlight that the strategies put forward in this study have attempted to address several of the current challenges in this research area and indicate a new way to face such problems. With regard to electoral predictions, we present a new approach for estimating voter preferences, which is able to complement traditional polling or even be incorporated into the polling methodology. For the ML area, the present study has investigated and applied ML for a new, as yet underexplored, context that presents very specific challenges. For the social media area, this study has examined the possibility of finding nonlinear correlations between online behavior and offline outcomes.

2. Problem definition and methodology

2.1. Main challenges addressed

The problem definition considers the open challenges presented in Section 1.1 and the challenges of the political scenario for using ML in the political and electoral contexts.

The political and electoral contexts vary widely between countries. The U.S. presidential elections are the most studied but present a very particular scenario, and the results are hard to replicate in other scenarios. Moreover, since presidential elections usually take place every four or five years, few historical data are available. Also, there is little party loyalty in some countries, as in Latin America, changes in the scenario often occur between elections, and the list of candidates may change even during the campaign. Thus, it is hard to obtain consistent historical data to train ML algorithms for election predictions. Indeed, there is almost no available data for ML training: historical data are scarce, and there is only one actual labeled data, the final vote share, which is the aim of prediction.

One direction for addressing this challenge is to use traditional polls as labeled data to train the models. However, the use of polls adds a new set of challenges. They present a variety of errors, and their results are often challenged. There are also many differences between pollsters, their methodology, presentation method, and results, which makes it challenging to group their results as a unique set of labeled data. Additionally, there are no evenly spaced time intervals between the polls, which usually decrease as election day approaches, thereby imposing constraints on using traditional time-series approaches.

2.2. Research questions and hypotheses

The main objective of this study is to define a process and create an ML model based on the SM performance of candidates, which is capable of performing daily nowcasting and providing final predictions of election results with competitive results to traditional polls. Thus, the following research questions were defined:

RQ1. Is it possible to define a process and create an ML model capable of predicting election results based on the SM performance of candidates?

RQ2. Is it possible to define a process and create an ML model capable of performing daily nowcasting of election results based on the SM performance of candidates?

To answer RQ1 and RQ2, this study considered a postpositivist stance (Easterbrook, Singer, Storey, & Damian, 2008). Thus, the following hypotheses were defined:

H1. It is possible to define a process and create a model based on the SM performance of candidates, using an ML approach trained with traditional polls, which is capable of predicting election results with competitive results to traditional polls.

H2. It is possible to define a process and create a model based on the SM performance of candidates, using an ML approach trained with traditional polls, which is capable of making daily predictions of election results with competitive results to traditional polls.

These hypotheses lead to their respective null hypotheses, which researchers aim to refute:

H1'. It is not possible to define a process and create a model based on the SM performance of candidates, using an ML approach trained with traditional polls, which is capable of predicting election results with competitive results to traditional polls.

H2'. It is not possible to define a process and create a model based on the SM performance of candidates, using an ML approach trained with traditional polls, which is capable of making daily predictions of election results with competitive results to traditional polls.

In order to reject the null hypotheses of H_1' and H_2' in favor of hypotheses H_1 and H_2, which answers the research questions, we defined a methodology with four main steps: (i) to define the concept of SM performance; (ii) to define a framework composed of a process and ML modeling for election prediction; (iii) to collect all the required data, including SM, polls, and final vote share; (iv) to run experiments by applying the framework on collected data for prediction; and (v) to

evaluate the results by comparing predictions with strong baselines.

The next section presents steps (i) and (ii), and Section 4 presents the experiments covering steps (iii) to (v).

3. The proposed framework

This section presents the concept of SM performance used in this study, SoMEN – the Social Media framework for Election Nowcasting, and SoMEN-DC, an execution strategy for SoMEN that enables continuous prediction during the campaign (DC).

3.1. Measuring social media performance

The measurement of the SM performance applied in this study uses the approach presented by Brito et al. (Brito et al., 2021), inspired by Zajonc's exposure theory (Zajonc, 1968), in conjunction with an understanding of how the algorithms on social media platforms function.

In 1968 (Zajonc, 1968) and beyond (Zajonc, 1980, 2001), Zajonc's studies hypothesized that "mere repeated exposure of the individual to a stimulus object enhances his attitude towards it." This effect, also termed the familiarity principle, has been demonstrated in many different contexts and it may be considered that people tend to have better attitudes towards others whom they are used to seeing. The principle is also valid for correlations between politicians' exposure and electoral performance (Oppenheimer, Stimson, & Waterman, 1986).

Considering this effect, the performance measurement focuses on analyzing the number of people who pay attention to the candidates by interacting with their content within the context of SM. For this, we considered the interactions on posts of the official profiles of candidates on newsfeed platforms, such as Facebook, Twitter, and Instagram. The interactions consist mainly of the likes, comments, and shares (or similar terms, such as retweets on Twitter as a synonym for shares) on each post. These actions indicate that the user has seen and paid attention to the content and has actively reacted.

All these actions help to propagate a candidate's presence online. Since SM algorithms give priority to showing the content of users with more engagement (Lars Backstrom and The Facebook, 2013; Youtube Inc., 2017), this creates a snowball effect. As more people interact with a post, so it is shown to more people, leading to more people interacting with it. The end result of the exposure theory is that more engagement and more exposure may be correlated with a better attitude towards a candidate and more votes.

We have considered the absolute numbers of each interaction in a period and the relative numbers per post. These metrics are generic and well suited for most newsfeed-based SM platforms. In the specific case of this study, we consider Facebook, Twitter, and Instagram as SM platforms, as will be detailed in Section 4. However, if other relevant platforms are identified as being relevant in other election scenarios, these metrics may also be added by following the same rationale of interactions, even if they present slight differences. For example, in the case of YouTube, the number of visualizations, likes, dislikes and comments on the videos posted by the candidates should be considered. Table 2 summarizes the metrics used, the same as (Brito et al., 2021), who explained them in more detail.

Table 2
Metrics for performance measurement (Brito et al., 2021).

Social network	Metric	Description
Facebook	FBPosts	Sum of posts in the period
	FBLikes	Sum of likes in the period
	FBShares	Sum of shares in the period
	FBComments	Sum of comments in the period
	FBLikesPPost	Average of likes per post in the period
	FBSharesPPost	Average of shares per post in the period
	FBCommentsPPost	Average of comments per post in the period
Twitter	TTPosts	Sum of posts in the period
	TTLikes	Sum of likes in the period
	TTRetweets	Sum of retweets in the period
	TTLikesPPost	Average of likes per post in the period
	TTRetweetsPPost	Average of retweets per post in the period
Instagram	IGPosts	Sum of posts in the period
	IGLikes	Sum of likes in the period
	IGComments	Sum of comments in the period
	IGLikesPPost	Average of likes per post in the period
	IGCommentsPPost	Average of comments per post in the period

3.2. SoMEN – A Scial media framework for election Nowcasting

Predicting the final vote share of candidates is a regression problem, i.e. when the objective is to predict or estimate a numerical value. The training and predictions are based on the SM performance data as features. Polling data are used as imprecise labeled data for supervised training during a time prior to elections. Thus, the prediction attempts to closely match the election vote share, which has only one sample. Below, we present the defined process and the ML model.

The SoMEN process is based on CRISP-DM (Shearer, 2000), one of the best-known processes for data mining, in such a way that it is generic and may be adapted for presidential elections worldwide. The process contains five phases: (i) election understanding, (ii) data collection and understanding, (iii) data preparation, (iv) modeling and execution, and (v) evaluation, as illustrated in Fig. 1.

3.2.1. Election understanding

The first phase is to understand the election scenario, which involves the election timeline, the candidates, SM platforms, pollsters, and polls.

Politicians may begin mobilizing their voters long before the election period. Thus, it is important to define a reasonable date for initiating data collection, at least at the official beginning of the campaign. This decision must be taken in advance because some SM platforms may limit the gathering of past data. We therefore suggest a date between 6 and 10 months before elections, because this is normally when the list of official candidates starts to take form, and the initial polls start being published.

The list of probable candidates must be pruned until the election day, because the candidates may change even during the official campaign. Another reason for pruning is because candidates with low voting intentions may bias the predictions in two ways. First, polls usually group minor candidates in the "other" category, which does not allow models to be trained with their data. Second, prediction errors may lead to a misinterpretation of the results, since an error of 0.5 percentage points regarding a candidate with 30% of votes is small, but the same error regarding a candidate with 0.1%age points is remarkably high. Thus, we suggest that candidates should only be considered if they consistently present at least 1% of the voting intentions in the polls.

It is also necessary to identify the main SM platforms used in the country where the election will be held, and to find the profiles of the candidates on each platform, preferentially verified accounts.

Lastly, a decision must be taken on which pollsters and polls to use in the training set. Some countries, such as the U.S., have a high number of publicly available polls and daily weighted averages, such as (The Huffington Post, 2016), The New York Times (The New York Times, 2016), and Real Clear Politics (Real Clear Politics, 2016). Nevertheless, access to polls is a barrier in many countries. Thus, some strategies for pollster selection may be adopted, such as selecting the pollsters with the highest reputation and/or with the best results on previous elections, using results of poll aggregation sites, or including pollsters with a minimum number of polls. This decision must be taken carefully since it may directly affect the results.

3.2.2. Data collection and understanding

The second phase involves collecting two sets of data: SM data and poll data.

Collecting SM data is challenging both for social and ML researchers

Fig. 1. The SoMEN Process.

because it is difficult to find public datasets with this data. Thus, complete information systems must be developed or acquired, and pass through the platform's review process, as occurs with Facebook and Instagram.[1] After this, the candidates' posts, including the related metrics presented in Section 3.1, must be gathered daily from the API of the platforms. Other strategies consider scraping the data or using third-party applications, however, the completeness of data may not be assured.

Collecting poll data is also a challenge. The collection may be diverse and frequently depends on manual collection from the websites of the electoral court, as in Mexico and Colombia, directly on the pollster websites, or even from document repositories such as Scribd (scribd.com), as in Argentina.

The data understanding step should then be performed. With regard to SM data, it is necessary to understand how candidates are using the SM platforms so as to identify which platforms are the most commonly used and the most relevant, and which should be considered in the prediction model. It is also important to verify data distribution and whether it is suitable for the chosen model. An understanding of poll data is required in order to prune the data and select which polls should be used.

3.2.3. Data preparation

This phase aims to prepare the collected data for the prediction model.

Because SM data used in this approach is public data collected from the official profiles of politicians using official application programming interfaces (APIs), the data are complete, and data cleaning is not necessary. However, the initial dataset should be enhanced and transformed.

Data are modeled so that the result r of a poll (or election results) on a specific date d is a function of the SM performance observed in the candidate's profiles in an aggregate window of w days prior to d. In the specific case of this study, considering Facebook, Twitter, and Instagram as SM platforms, it is modeled as:

$$r = f(F, T, I)_{d-w \ldots d-1}$$

We use the 17 features presented in Table 2 for the SM performance. As an example, for a poll published on January 30 and considering a window of 28 days, input data are the individualized sum of all the posts, likes, comments, and shares/retweets from Facebook, Twitter, and Instagram from January 2 to January 29, and the ratio "per post" per platform for all of them.

The definition of the window size, i.e., the number of collection days before the target data, is central to data preparation and, most often, taken arbitrarily, as discussed in Section 1.1. However, in previous studies (Brito et al., 2021), no correlations were found between the window size and the success of predictions. Additionally, we must consider that the electorate's behavior is not uniform, since some people may access the SM platforms daily, while others access the platforms at different intervals. Thus, we adopted a strategy based on generating different datasets with different window sizes and creating a committee machine through an ensemble of estimators, each using one different dataset related to a different window, and averaging the results. This ensemble of estimators may be better suited for capturing different user behavior and thereby avoiding the bias of selecting an arbitrary window size.

The small number of available polls in many elections leads to a small number of data samples for training and may lead to the well-recognized problems of violation of the VC-dimension (Vapnik, Levin, & Le Cun, 1994). Thus, it is desirable to use feature selection or dimensionality reduction techniques, such as principal component analysis (PCA). In this context, the PCA is well suited because it eliminates collinearity amongst features, which is likely in this scenario, while allowing dimensionality reduction.

3.2.4. Modeling

This phase aims to choose an appropriate ML model and design an appropriate architecture for the predictions.

The problem of predicting the candidate vote share is characterized as a regression problem because many continuous values are predicted, the vote shares.

In order to choose an appropriate method, the small sample size of this domain, based on the number of available public polls, must be considered. The method should also be generalizable and should not depend on assumptions regarding the distribution of input data. Moreover, due to the low success of linear methods, the model must be capable of drawing nonlinear mappings.

One challenge of the ML methods involves tuning its parameters. Hence, it is desirable to choose parameters by selecting similar problems in the literature or using techniques for automatically selecting parameters, such as grid search. The main disadvantage of the last approach is the computational power needed to perform all the calculations of all the parameter combinations. One way to reduce the computational power needed to find the best parameters is to use a randomized search, which randomly chooses some parameters from the grid search. It has been demonstrated that this approach may provide similar results to grid search, demanding less computational power (Bergstra & Bengio, 2012).

3.2.5. Evaluation

Prediction evaluation is a challenge for the polling industry. Results must be measured with relevant metrics for the domain and compared with well-defined baselines, and the evaluation must measure the difference between the predicted results and the candidate's final vote share. It must also compare the errors obtained by predictions with the errors obtained by polls.

Two metrics defined after the 1948 polling crisis, the "Mosteller Measure 3" and "Mosteller Measure 5" (Mosteller, Hyman, McCarthy, Marks, & Truman, 1949) have been widely used ever since in order to evaluate the accuracy of elections polls (Hillygus, 2011; Jennings & Wlezien, 2018; Martin, 2005). The first is the average absolute error between the prediction and the actual results of all candidates. In statistics, it is widely known as the mean absolute error (MAE). The second, which we call the absolute error on the margin (AEOM), is the absolute value of the difference between the margin separating the two leading candidates in the poll and the actual vote.

Other support metrics may also be used, such as the mean absolute percentage error (MAPE), which measures the percentage error, and the root mean squared error (RMSE). The MAPE is relevant because, for example, an error of 3 points in a vote share is much more relevant for a candidate with 2% of votes than for a candidate with 50% of votes, and the MAE does not capture this relevance. Moreover, the RMSE may also

[1] https://developers.facebook.com/docs/app-review

expose outliers.

Thus, while the primary comparison metric is the MAE, these other metrics should be used in order to give more significance to the results.

The errors obtained by predictions should be compared with two baselines. The first is the historical MAE threshold, and the second is the errors obtained by the polls used for training the model. Jennings and Wlezien (Jennings & Wlezien, 2018) analyzed more than 30,000 national polls from elections in 45 countries between 1942 and 2017 and found an MAE of 2.70 for presidential elections (standard deviation of 2.13), with almost no variation over the years. Thus, errors below the historical MAE or within one standard deviation (equivalent to 68% of historical values) may be considered in line with historical data. In addition, the errors will be compared with those obtained by the last prediction of traditional polls individually, and with the final poll average.

Lastly, although rarely found in related studies, the prediction errors should be statistically tested with the errors obtained by polls to verify whether they are statistically equivalent, higher or lower. For statistical tests, the Wilcoxon signed rank test (Wilcoxon, 1945) should be used because it is a scenario of matched samples and needs a non-parametric test because we cannot assume a normal distribution of sample means.

3.2.6. The SoMEN ML model

We propose a specific ML model and an architectural design. It is important to note that many different choices may be made, and the objective is not to pursue the best model but rather one that is suitable and reasonable for this context.

Thus, for this modeling, the following decisions were taken:

- To reduce issues related to volume manipulation, such as the existence of astroturfing, spam, paid propaganda, and the use of BOTs, each candidate will be trained and predicted individually. Thus, the models are trained with the specific behavior of the supporters of each candidate;
- To reduce the dimensionality, as well as to allow the use of high-correlated features present on SM performance features, a PCA will be performed on the input dataset;
- Ten different datasets will be generated, with various windows sizes: 1.0.7, 14, 21, and 28 days;
- A committee machine composed of an ensemble of 10 predictors will be used: each will receive one dataset related to a specific window size, and the prediction will be the average of the members' predictions;
- The preferred ML methods for prediction are the MLP-BP and the GRNN.

The Multilayer Perceptrons (MLP) and the general regression neural networks (GRNN) are sophisticated models used for regression. They are of the class of Artificial Neural Networks (ANN), a computational model inspired by how the human brain functions and designed to solve problems that traditional computing does not perform well, including perception problems such as face and speech recognition (Haykin, 1998). The main characteristics of ANNs include their highly distributed nature and the ability to "learn" from previous examples.

The MLP is one of the most popular neural network architectures. In brief, it is based on an architecture of connected units with weights, that may be trained in a supervised training approach (by trial-and-error) and is capable of being applied in a new stimulus which has not been seen before (Haykin, 1998). The MLP-BP, a MLP trained with back-propagation strategy, was chosen because of its inherent characteristics, such as non-linearity, there is no need for assumptions on the distribution of input data, it has good generalization capabilities, and it performs well even with the existence of noise data, plus it is proven to be a universal approximator (Hornik, Stinchcombe, & White, 1989). The choice is also based on a recent extensive experimental survey that compared 77 popular regression methods (Fernández-Delgado et al., 2019). In the study, the MLP-BP based model designed with one hidden layer and a committee of 5 identical MLP-BPs trained using different random seeds, obtained remarkable results with small datasets.

The GRNN (Specht, 1991) also uses the supervised training approach, and falls into the category of probabilistic neural networks (PNN) (Specht, 1990). The use of probabilistic neural networks is especially advantageous because the network learns in one pass through the data and is able to generalize from examples the moment they have been stored (Specht, 1991). Thus, it presents a number of characteristics well suited to this context. It is particularly advantageous with small sample data and needs few examples in order to achieve comparable accuracies with a MLP-BP model (Specht, 1991). Moreover, it only requires one hyperparameter to be set, a distinguishing difference from the MLP. It also converges to global minima, and is quicker to train, despite being lower for predictions.

In summary, Fig. 2 illustrates the SoMEN instantiation, considering the data preparation and the modeling phases.

3.3. SoMEN-DC: A social media framework for election Nowcasting during the campaign

The SoMEN-DC is an execution strategy for SoMEN to continuously repeat the process so that daily predictions may be made.

The execution consists of (i) Gathering an initial number of polls and the SM performance data; (ii) Training the model with available data; (iii) Collecting SM data daily and making out the sample, daily predictions; and (iv) Retraining the model with a new set of labeled data when new poll data is released. Fig. 3 illustrates the steps.

The first decision concerns the minimal number of polls needed to train the model and to begin making predictions. It is expected that when more poll data arrives and, consequently, more labeled data is available, prediction accuracy increases. This usually occurs in the poll domain: more accurate predictions are made as the election date approaches.

The second decision concerns retraining the model. The previously chosen models, MLP-BP and GRNN, are well suited for this context. However, other approaches may be better suited, such as those based on online learning (Losing, Hammer, & Wersing, 2018), because they may perform incremental learning, and may be promising as future research.

Evaluating daily predictions is even more challenging than evaluating the final predictions. We may test the final prediction with the final vote share, but for daily predictions we only have an imprecise measure of it, the poll data. As a consequence, it is difficult to obtain a conclusive, statistically demonstrated result.

Thus, the approach of comparing predicted results, polls, and the final vote share seems to be the most plausible option, although statistical tests on results may lead to wrong conclusions. Thus, we should perform two analyses. The first is a descriptive, qualitative analysis of the two most voted candidates regarding polls, predictions, and the final vote share. The second considers the measurement of prediction errors by considering polls as imprecise ground truth. Hence, competitive predictions will be considered as those that, compared with polls, present errors below the historical MAE of 2.7 percentage points, within 3.00 deviation, i.e., the error margin considered for most polls. Indeed, this approach may also be seen as a cross-validation approach of the SOMEN, using a strategy of leave-future-out (Bürkner, Gabry, & Vehtari, 2020), a variation of the well-established leave-one-out strategy.

4. Experiments

We performed experiments on predicting recent presidential elections in Latin America. Countries throughout Latin America present a similar context: they are situated in the same region, they have similar historical origins, languages, characteristics and electoral procedures, such as the direct vote, the two-round system, many candidates running in the first round, and few available polls. Thus, to apply the same

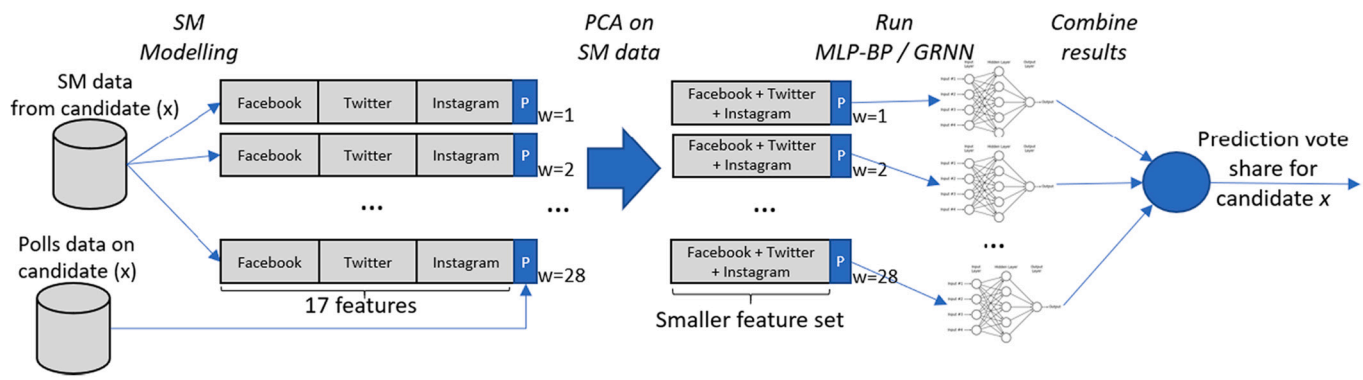

Fig. 2. SoMEN instantiation.

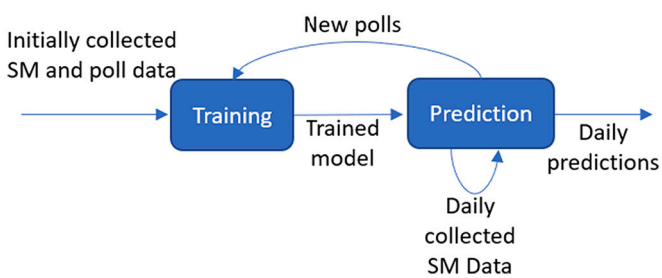

Fig. 3. The SoMEN-DC execution.

methodology to these countries enables an evaluation of the framework and a comparison of results.

Presidential elections held in Colombia, Mexico, and Brazil in 2018 and Argentina in 2019 were chosen for the experiment. These are the four most populous countries in the region, home to 70% of the population (United Nations, 2019), and were responsible for 81% of the GDP in 2019,[2] which increases the validity of our experiments. We only considered the first round of elections, because the election context is different to the second round: it involves many candidates with varying strategies, because there is unequal exposure on TV and in the traditional media.

4.1. The SoMEN execution

4.1.1. Election understanding

The selected elections took place in 2018 and 2019, and the official campaigns lasted between two to four months. In all countries, except Brazil, polls may not be released in the week before the election date. To maximize the amount of collected data, mainly by considering that there are few available polls in the selected countries, data collection started 10 months before the elections (300 days), when the candidatures start to take form.

There were 6 candidates in Argentina, 13 in Brazil, 8 in Colombia and 4 in Mexico. To avoid any bias generated by candidates with small vote intentions, the study considered only candidates with more than 1% of vote intentions, which resulted in all 6 candidates in Argentina, 5 in Brazil and Colombia, and all 4 in Mexico.

At the time of elections, the most used newsfeed-based SM media platforms in these countries were YouTube, Facebook, Instagram, and Twitter (Kemp, 2019a, 2019b, 2019c; Kemp, 2020). YouTube data were not considered because we identified that at the time of the elections, many candidates had no official profile on this platform, or its use was irregular. Also, the Instagram profiles of two Mexican candidates (Obrador and Cortés) were personal profiles, rather than commercial profiles. Because this setup does not allow for data to be collected through the Instagram API, data on these two candidates on this platform were not included, but data from the other two (Kuribeña and Calderon) was used.

Identifying pollsters was a manual task, and was performed on three sources, in this order: (1) On the official electoral court website, (2) On the Wikipedia entry describing the election in the official language of the country (Portuguese or Spanish) and in English, and (3) through an open search on Google using the words "*pesquisa eleições 2018 Brasil*" for the Brazilian elections, and "*encuestas electorales* < *country* > <*year* > "for the other countries, replacing <*country* > for the country name and < *year* > for the year of election. The findings and pruning of the list of pollsters and polls is detailed in the following section.

4.1.2. Data collection and understanding

4.1.2.1. Data from SM platforms. Data collection was made by the official API using a set of tools previously developed by the authors and used in earlier studies. The dataset used in this study is the same collected and used by (Brito & Adeodato, 2020) (Brito & Adeodato, 2022).

A total of 67,403 posts and 449,201,837 interactions were collected from the candidate profiles in all four countries, thereby constituting an average of 6664 interactions per post. A total of 15,432 posts was collected from the Argentinian candidates: 3224 from Facebook, 10,444 from Twitter, and 1764 from Instagram. A total of 19,586 posts was collected from the Brazilian candidates: 6101 from Facebook, 10,181 from Twitter, and 3304 from Instagram. A total of 22,542 posts was collected from the Colombian candidates: 3746 from Facebook, 15,996 from Twitter and 2800 from Instagram. Lastly, a total of 9843 posts was

[2] https://data.worldbank.org/indicator/NY.GDP.MKTP.CD?locations=ZJ&most_recent_value_desc=true Viewed on January 01, 2021.

Table 3
The pollsters and number of polls collected for each country.

Country	Pollster	Polls
Argentina	Federico González & Asociados	11
	Ricardo Rouvier & Asociados	8
	Gustavo Córdoba y Asociados	7
	Circuitos	5
Brazil	Ipespe	19
	Datafolha	11
	IBOPE	10
	DataPoder360	6
	FSB	6
	MDA	5
	Paraná Pesquisas	5
Colombia	Centro Nacional de Consultoria	8
	Cifras Y Conceptos	7
	Datexco Company	6
	Guarumo, Ecoanalítica	6
	Invamer Sas	6
	Yanhaas	6
Mexico	Parametria	9
	Grupo Impacto	8
	Arias Consultores	7
	Economista y Asociados	7
	El Financiero	7
	Suasor Consultores	7
	Conteo	6

collected from the Mexican candidates: 2308 from Facebook, 7146 from Twitter, and only 389 from Instagram.

Detailed analysis of the dataset may be found in (Brito & Adeodato, 2022). To summarize, candidates in all countries mostly posted on Twitter, but the public response on this platform is lower than the response on Facebook and Instagram. In Argentina and Brazil, most interactions were on Instagram, while in Colombia, most interactions were on Facebook. Moreover, there is a large variation in the interactions on posts since they received between zero and one million interactions, and there is a large standard deviation in all metrics.

4.1.2.2. Data from polls. Polls were found on the following sources: Mexican[3] and Colombian[4] polls from the electoral court websites, Argentinian polls from pollster websites and on the Scribd document repository (scribd.com), and Brazilian polls from a website aggregator, Poder 360.[5] With the exception of data from Brazil, data for each survey was collected manually, including data referring to methodology.

Consideration was given to: (i) pollsters which had performed at least 5 polls, due to consistency and continuity; (ii) pollsters with the last poll being performed at least 15 days before the election (or limit) date, to enable a comparison of the last poll with the election result; and (iii) national polls, performed with a broad public, for president in the first round. Polls performed only on the internet on platforms based on self-selection were not included, to reinforce the third criteria, polls performed on a broad public, and to avoid reintroducing the bias related to the non-representativity of the population on the internet and on the SM platforms. Table 3 presents the pollsters and the number of polls considered after poll selection and pruning.

Table 4 presents a summary of the last poll performed by each pollster before the election. The lowest standard deviation was observed in Argentina, signifying that the final polls somehow agreed, and that the predictions converged. On the other hand, the final predictions from Mexico were very different, ranging from 34.0 to 62.5 percentage points for the most voted candidate. Also, the poll average indicated a wrong list of the most voted candidates, since it indicated Kuribreña as the second most voted candidate, who in fact was in third place.

Table 5 presents a summary of all poll data, which demonstrated high variations. This, however, was expected since voting intentions may vary throughout the campaign.

All poll-collected data, including the official documents, methodology, and values, is available at *kellyton.com.br/somen*.

4.1.3. Data preparation

A set of 10 independent datasets was generated for each of the considered candidates, with the features described in Section 3.1, and the target was the vote share presented by the polls. Each dataset was generated with a different window size, $\boldsymbol{w} = [1.0.7, 14, 21, 28]$ days. Each sample is based on a poll day, considered as being one day after the last date of interviews, found in the poll methodologies, and considered the performance of the candidate on the SM platforms during $\boldsymbol{w}$ days before the poll day. An example of the dataset is presented on the Appendix.

One additional sample was also generated, correlating SM features and the final vote share instead of the poll data. This sample was not used in the training, although the features were used for an out-of-sample prediction.

In order to generate the final datasets to be used for training and predictions, the features were normalized and as planned, due to the few training samples (from 31 in Argentina to 62 in Brazil) for the number of features (17), PCA was applied to each dataset independently. Component selection was set to cover a variance higher than 95%, a well-accepted level in the research literature. Thus, the number of components varied from two to seven, thereby drastically reducing the number of input features.

4.1.4. Modeling

As already defined, the MLP-BP and GRNN were used. A baseline linear regression model was also used.

For the MLP-BP model, to avoid the well-known problem of local minima, a committee was created of 5 identical MLP-BPs trained using different random seeds, the results of which were averaged to give a unique output. For parameter selection, we considered the results obtained by (Brito & Adeodato, 2020) in a previous, similar study. Their results demonstrated that although the grid search parameters approach increased the execution time by 240, it did not increase the prediction accuracy when compared with following context-based manually selected parameters: one hidden layer with three neurons, to avoid overfitting; L-FBGS as a solver, which performs well with small samples; an alpha set to 0.05 for fast convergence, a constant learning rate for fast training, and logistic activation. Thus, the fixed parameter approach, with the abovementioned parameters, was used.

In the GRNN model, there is only one hyperparameter to be adjusted, the smoothing parameter. As no baseline value was found for this parameter, it was found by the grid search approach, varying from 0.1 to 4, in steps of 0.2.

As planned, a committee machine was created for each model, composed of an ensemble of 10 predictors. As input, each received a different dataset related to a different aggregated window size, and all predictions were averaged as the final prediction for that candidate. The machines were trained with all the available poll data until one day before the elections, and one prediction was made for the final vote share. The results were then compared with the actual vote share of each candidate.

To compare and verify the gain obtained by applying PCA, experiments were applied twice: with and without the application of PCA. To summarize, six experiments were undertaken: linear regression, linear regression with PCA, MLP-BP, MLP-BP with PCA, GRNN, and GRNN with PCA.

[3] https://computos2018.ine.mx/#/presidencia/nacional/1/1/1/1. Viewed in October, 2020.

[4] https://www.cne.gov.co/inventario-de-encuestas. Viewed in October, 2020.

[5] https://www.poder360.com.br/banco-de-dados/. Viewed in October, 2020.

Table 4
Summary of the final week of polls.

Country	Candidate	Period	Min	Max	Mean	Median	Std. Dev.
Argentina	Fernandez	Oct. 10–18	49.50	50.90	50.15	50.10	0.60
Argentina	Macri	Oct. 10–18	29.20	32.50	31.33	31.80	1.45
Argentina	Lavagna	Oct. 10–18	7.20	9.20	7.83	7.45	0.93
Argentina	del Caño	Oct. 10–18	2.20	2.80	2.53	2.55	0.28
Argentina	Centurión	Oct. 10–18	1.20	1.70	1.48	1.50	0.21
Argentina	Espert	Oct. 10–18	0.80	1.90	1.28	1.20	0.46
Brazil	Bolsonaro	Sep. 30 - Oct.07	30.00	36.70	34.37	36.00	2.71
Brazil	Haddad	Sep. 30 - Oct.07	21.80	25.00	22.97	22.00	1.32
Brazil	Gomes	Sep. 30 - Oct.07	9.00	15.00	11.19	11.00	2.14
Brazil	Alckimin	Sep. 30 - Oct.07	5.80	11.00	7.46	7.00	1.64
Brazil	Amoêdo	Sep. 30 - Oct.07	0.00	5.00	2.63	3.00	1.50
Colombia	Márquez	May 15–18	35.00	41.50	36.92	36.20	2.45
Colombia	Petro	May 15–18	24.00	34.80	27.58	26.50	4.07
Colombia	Fajardo	May 15–18	14.00	18.00	16.05	16.00	1.27
Colombia	Lleras	May 15–18	6.00	14.30	9.02	8.30	3.34
Colombia	Calle	May 15–18	1.90	4.00	2.80	2.65	0.84
Mexico	Obrador	Jun. 18–26	34.00	62.50	41.12	38.34	8.74
Mexico	Cortés	Jun. 18–26	14.00	24.00	19.67	21.00	3.82
Mexico	Kuribreña	Jun. 18–26	13.00	29.00	19.74	15.77	6.17
Mexico	Calderon	Jun. 18–26	1.00	9.00	3.77	2.49	2.71

* The date of the poll was considered as one day after the last day of interviews.

Table 5
Summary of all polls.

Country	Candidate	Period	Min	Max	Mean	Median	StdDev
Argentina	Fernandez	Jan. 12 - Oct. 18	35.70	54.50	45.07	48.10	6.11
Argentina	Macri	Jan. 12 - Oct. 18	22.00	35.20	29.32	29.91	3.72
Argentina	Lavagna	Jan. 12 - Oct. 18	5.70	19.30	9.56	8.90	3.11
Argentina	del Caño	Jan. 12 - Oct. 18	0.40	4.97	3.18	3.05	1.09
Argentina	Centurión	Jan. 12 - Oct. 18	0.70	3.00	1.48	1.40	0.62
Argentina	Espert	Jan. 12 - Oct. 18	0.80	5.82	2.71	2.33	1.57
Brazil	Bolsonaro	Jan. 31 - Oct.07	17.00	36.70	25.60	24.00	5.29
Brazil	Haddad	Jan. 31 - Oct.07	1.00	25.20	11.04	7.00	8.67
Brazil	Gomes	Jan. 31 - Oct.07	8.00	15.00	10.57	10.45	1.60
Brazil	Alckimin	Jan. 31 - Oct.07	5.30	11.00	7.98	8.00	1.18
Brazil	Amoêdo	Jan. 31 - Oct.07	0.00	5.00	2.50	3.00	1.08
Colombia	Márquez	Jul. 31 - May 18	0.40	45.90	22.85	23.60	15.97
Colombia	Petro	Jul. 31 - May 18	9.00	34.80	20.73	22.00	7.17
Colombia	Fajardo	Jul. 31 - May 18	5.00	26.00	13.93	14.00	4.33
Colombia	Lleras	Jul. 31 - May 18	5.30	15.60	8.49	7.90	2.53
Colombia	Calle	Jul. 31 - May 18	1.00	11.00	4.29	4.00	2.38
Mexico	Obrador	Sep. 20 - Jun. 26	20.80	69.90	36.63	34.90	9.82
Mexico	Cortés	Sep. 20 - Jun. 26	12.00	28.00	20.18	21.00	3.58
Mexico	Kuribreña	Sep. 20 - Jun. 26	11.00	29.00	18.61	16.95	4.97
Mexico	Calderon	Sep. 20 - Jun. 26	0.60	9.00	2.61	2.00	1.84

4.1.5. Evaluation

The evaluation was performed according to the definitions in Section 3.2.5, with a comparison of the predictions, the actual results, prediction errors, and poll errors. Statistical tests were also performed and are detailed in the following section.

All experiments were run in a standard laptop computer with the following configuration: Processor Intel Core i7-8550U, RAM 16GB, data on an SDD 128GB disk. The implementation was in python using Scikit-learn (Pedregosa et al., 2011) for most computations and pyGRNN[6] for GRNN implementation. All the outputs, including detailed outputs for each window size, are available at *kellyton.com.br/somen*.

4.2. The SoMEN-DC execution

The execution of the SoMEN-DC was based on the SoMEN execution. The main difference is that instead of predicting only the final results, it made daily predictions, using the available data until one day before. The execution strategy consisted of the following stages:

1. Initial setup: first training with 10 polls with data on all candidates;
2. Begin daily predictions after the initial setup, considering SM and poll data until D-1;
3. Out-of-sample predictions: the prediction of an arbitrary day D considers polls before D;
4. When new poll data is available, retrain the model with the inclusion of this new data.
5. Error metrics are based on predictions on days with available poll data.

The evaluation was performed according to the definitions in Section 3.3. It may also be seen as a cross-validation approach of the SOMEN using a strategy of leave-future-out (Bürkner et al., 2020), thus reinforcing the validity of the study.

[6] Available at: https://github.com/federhub/pyGRNN. Viewed on: October 5, 2020.

Table 6
Predictions for the Argentinian elections.

Candidate	Election result	Polls Average	Predictions					
			LR	LR PCA	MLP	MLP PCA	GRNN	GRNN PCA
Fernandez	47.02	50.15	53.27	51.73	47.44	47.00	47.41	46.82
Macri	39.27	31.33	43.07	34.39	31.14	31.14	30.70	30.66
Lavagna	5.99	7.83	8.74	8.25	7.68	7.75	7.95	7.96
Caño	2.10	2.53	3.95	2.59	2.70	2.33	2.76	2.77
Centurión	1.66	1.48	1.89	1.34	1.31	1.20	0.94	0.92
Espert	1.43	1.28	7.07	2.41	2.78	2.79	1.86	1.91

Table 7
Predictions for the Brazilian elections.

Candidate	Election result	Polls Average	Predictions					
			LR	LR PCA	MLP	MLP PCA	GRNN	GRNN PCA
Bolsonaro	32.63	34.37	37.90	34.69	30.65	32.13	34.57	34.56
Haddad	20.75	22.97	36.37	30.76	20.68	20.93	23.77	23.78
Gomes	8.84	11.19	14.24	11.38	10.83	10.85	10.04	10.00
Alckmin	3.37	7.46	6.14	7.26	7.27	7.04	7.20	7.53
Amoêdo	1.77	2.63	2.25	2.63	2.87	2.58	2.41	2.44

Table 8
Predictions for the Colombian elections.

Candidate	Election result	Polls Average	Predictions					
			LR	LR PCA	MLP	MLP PCA	GRNN	GRNN PCA
Márquez	39.34	36.92	61.59	56.71	36.73	37.85	38.29	38.43
Petro	25.08	27.58	36.38	28.87	26.36	26.41	27.16	27.13
Fajardo	23.78	16.05	64.85	14.22	12.52	13.65	15.57	15.53
Lleras	7.3	9.02	7.32	7.95	8.52	8.54	8.89	8.84
Calle	2.05	2.80	4.64	4.44	2.85	3.01	2.61	2.58

Table 9
Predictions for the Mexican elections.

Candidate	Election result	Polls Average	Predictions					
			LR	LR PCA	MLP	MLP PCA	GRNN	GRNN PCA
Obrador	53.19	41.12	40.58	37.96	36.74	37.66	38.87	39.06
Cortés	22.28	19.67	18.36	20.10	19.42	20.20	20.21	20.21
Kuribreña	16.41	19.74	18.46	17.92	17.55	17.31	18.34	18.36
Calderon	5.23	3.77	6.75	3.66	2.85	3.00	2.93	2.90

Table 10
Error metrics for the Argentinian predictions.

Argentina - Error comparisons	MAE	MAPE	RMSE	AEOM
Ricardo Rouvier & Asociados	1.89	0.15	3.00	9.25
MLP-BP PCA	1.99	0.31	3.45	8.11
MLP-BP	2.09	0.32	3.45	8.55
GRNN PCA	2.11	0.28	3.64	8.41
GRNN	2.12	0.27	3.62	8.96
LinearRegression PCA	2.27	0.29	2.96	9.59
Poll Average	*2.28*	*0.17*	*3.57*	*11.08*
Circuitos	2.41	0.26	3.50	11.35
Gustavo Córdoba y Asociados	2.46	0.21	3.58	10.75
Federico González & Asociados	2.53	0.16	4.33	12.95
LinearRegression	3.42	0.94	4.01	2.45

Table 11
Error metrics from the Brazilian predictions.

Brazil - Error comparisons	MAE	MAPE	RMSE	AEOM
MLP-BP PCA	1.43	0.36	1.92	0.68
MLP-BP	1.81	0.41	2.20	1.91
Paraná Pesquisas	1.85	0.43	2.22	1.22
Ibope	2.13	0.32	2.48	2.12
GRNN	2.13	0.37	2.42	1.08
GRNN PCA	2.19	0.39	2.53	1.10
Poll Average	2.25	0.43	2.49	0.48
MDA	2.27	0.28	2.62	0.82
Ipespe	2.33	0.44	2.54	2.12
Datafolha	2.73	0.48	3.00	2.12
FSB	3.18	0.86	4.05	4.88
Datapoder 360	3.69	0.61	3.98	6.88
LinearRegression PCA	3.87	0.49	5.04	7.95
LinearRegression	5.91	0.52	7.86	10.35

Table 12
Error metrics from the Colombian predictions.

Colombia - Error comparisons	MAE	MAPE	RMSE	AEOM
GRNN PCA	2.66	0.18	3.89	2.96
GRNN	2.70	0.19	3.89	3.13
Guarumo, Ecoanalítica	2.87	0.21	3.95	0.86
Invamer SAS	2.98	0.14	4.02	2.26
Poll Average	3.02	0.22	3.88	4.92
MLP-BP PCA	3.03	0.23	4.67	2.82
Centro Nacional de Consultoria	3.14	0.35	3.45	5.26
MLP-BP	3.43	0.23	5.24	3.89
Yanhaas	3.46	0.24	4.86	5.26
Datexto Company	4.32	0.19	5.74	12.66
Cifras y Conceptos	4.33	0.43	5.13	3.26
LinearRegression PCA	6.75	0.45	9.10	13.58
LinearRegression	15.45	0.80	21.52	10.95

4.3. Experiment results – Research question 1

After running all the phases of the process, we attempted to refute H_1'. As planned, we compared our predictions with the actual results, the MAE error of our predictions with the historical threshold of 2.7 (standard deviation of 2.13), and also with the errors obtained by each pollster and by the poll average. We then performed a statistical test, by comparing the predicted errors with the errors obtained by the poll average. For all the measures, we considered raw results, since calculating the relative percentage may bias the results: mobilization on the candidate profiles was low at the beginning of the campaign, as was the number of people who had decided upon their vote. Thus, the sum would not reach 100%.

In Argentina, our predictions were performed with data trained until 10 days before the elections, the date of the last poll. In Brazil, the last poll occurs on the day before the elections. In Colombia and Mexico, the last poll is one week before the elections.

Tables 6 to 9 present the predictions and the final vote share, as well as the poll average. Tables 10 to 13 present the error metrics, including each pollster individually and the poll average.

Analyzing the Argentinian results, one pollster obtained a remarkable result, and all the MLP-BP and GRNN models obtained better MAE results than the other pollsters, and were also better than the poll average. With the exception of the linear regression, all predictions were under the historical MAE threshold of 2.70.

In terms of the Brazilian results, the MLP models obtained the best results. The prediction errors for the two most voted candidates using the MLP-BP PCA model were less than or equal to 0.5 percentage points. The GRNN models also obtained errors lower than the poll average and most of the pollsters, and all results generated by the MLP and GRNN models were under the MAE threshold of 2.70, although only four of the 7 pollster results were under this threshold.

Table 13
Error metrics from the Mexican predictions.

Mexico - Error comparisons	MAE	MAPE	RMSE	AEOM
Parametria	3.41	0.13	4.48	4.91
Reforma	3.59	0.19	5.61	10.99
*Poll Average**	*4.87*	*0.21*	*6.44*	*9.46*
Arias Consultores*	4.89	0.19	6.27	17.59
LinearRegression	5.03	0.21	6.72	8.69
GRNN PCA	5.12	0.23	7.3	12.06
LinearRegression PCA	5.12	0.19	7.77	13.05
GRNN	5.16	0.23	7.39	12.25
MLP-BP PCA	5.18	0.22	7.93	13.45
MLP-BP	5.71	0.24	8.45	13.59
Suasor Consultores*	6.43	0.27	8.75	15.91
El Financiero*	6.53	0.31	8.44	7.48
Economista y Asociados	6.83	0.35	9.65	18.71
Grupo Impacto	6.91	0.37	8.16	7.91
Conteo*	8.43	0.43	10.53	18.91
Pop Group*	9.32	0.50	11.69	17.91

Table 14
MAE errors of all countries, compared with the respective poll average.

MAE comparison	Argentina	Brasil	Colombia	Mexico	Average
MLP-BP PCA	1.99	1.43	3.03	5.18	2.91
GRNN PCA	2.11	2.19	2.66	5.12	3.02
GRNN	2.12	2.13	2.70	5.16	3.03
Polls Average	*2.28*	*2.25*	*3.02*	*4.87*	*3.11*
MLP-BP	2.09	1.81	3.43	5.71	3.26
Linear Regression PCA	2.27	3.87	6.75	5.12	4.50
Linear Regression	3.42	5.91	15.45	5.03	7.45

In Colombia, the GRNN models obtained the best results and were the only results equal to or below the historical threshold of 2.7. However, all results from the MLP-BP with and without PCA were within one historical standard deviation (4.83), as well as the pollsters. The MAE with the MLP-BP model with PCA were better than four of the six pollsters, very close to the MAE of the poll average. As in the previous countries, the linear regression models obtained the poorest results.

In Mexico, poll and prediction errors were higher than in the other countries. None of the polls or predictions achieved an MAE even close to the threshold of 2.7, and only two of the nine pollsters achieved errors within one historical standard deviation (4.83). Some pollsters presented an MAE higher than 7.0, which is unacceptable for this context. Also, despite a final difference of 5.87 percentage points between the second and third most voted candidates, most of the final polls and the poll average missed the ranking of candidates. The polls marked in the table with a star indicated the third most voted candidate, Kuribreña, as being the second most voted.

This result, including the high error in our predictions, was expected because of the wide variation in the final polls, ranging from 34.0 to 62.5 for the most voted candidate. Thus, the results were in line with the polls: the MAE of all predictions were poorer than 3 pollsters and the poll average, and better than 6 pollsters. Surprisingly, the linear regression model obtained the best results amongst the models, although it is difficult to identify why, due to the high variation of polls. Lastly, we highlight that the Instagram data of the two most voted candidates could not be collected, as shown in Section 4.1.1. Thus, if we had obtained access to this data, different results would have been obtained.

Table 14 summarizes the MAE obtained by the models in all countries, allowing a comparison with the poll average of each country. On average, the MLP-BP model with PCA obtained the best results. The two GRNN models also obtained better results than the poll average, and the MLP-BP model without PCA obtained competitive results. In all countries, the MLP-BP with PCA obtained better results than without PCA. The GRNN model with PCA obtained better results than the version without PCA in three of the four countries. It is worth noting that Mexico was the country with the second most polls (51), but obtained the worst results. This indicates that the quality of polls for training may be more relevant than the quantity.

We also performed Wilcoxon signed-rank tests. For this, we worked

Table 15
Wilcoxon signed rank test between the errors obtained with the prediction and polls.

Comparison	Two sided	Greater	Lower
Polls Average x Linear Regression	**0.02**	0.99	**0.01**
Polls Average x Linear Regression PCA	0.09	0.95	**0.05**
Polls Average x MLP-BP	0.75	0.62	0.38
Polls Average x MLP-BP PCA	0.28	0.14	0.86
Polls Average x GRNN	0.90	0.45	0.55
Polls Average x GRNN PCA	0.99	0.51	0.49

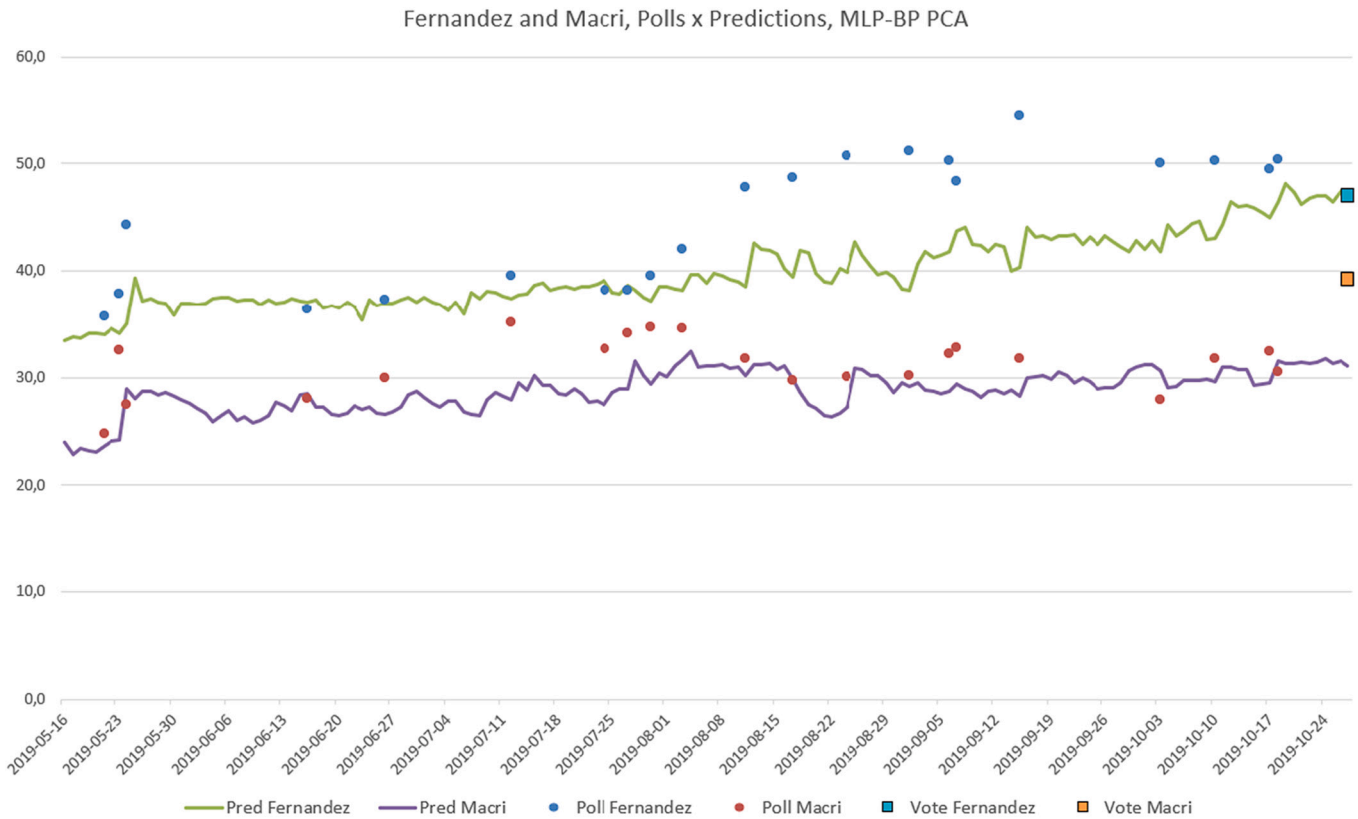

Fig. 4. Polls and predictions in Argentina: Fernandez and Macri, predictions using the MLP-BP PCA model and the final vote share.

with seven series, one for each prediction model and one additional with the poll averages. Due to the low number of samples, we concatenated the values of each country in series containing 20 points, related to the predictions of the candidates from all countries: 6 from Argentina, 5 from Brazil, 5 from Colombia and 4 from Mexico. Thus, we obtained more data in order to perform better statistical tests considering all predictions at once.

By comparing the absolute errors of our predictions and the poll averages, we obtained the *p*-values presented in Table 15. This demonstrates that the poll averages presented lower errors than both the linear regression approaches, although all other errors were similar. We highlight that besides the linear regression, the most significant result is $p = 0.14$, indicating that the errors of the poll averages may be higher than the errors obtained with the MLP-BP PCA approach, although the *p*-value is not strong enough to confirm this.

Considering the traditional and most used error metric in this scenario, the presented data illustrates that in all countries the results obtained with the two proposed models, the MLP-BP and the GRNN presented better or competitive results with the poll results. Moreover, the statistical analysis demonstrates that, statistically, the predicted results are competitive with the poll results. The presented data therefore **rejects** the null hypothesis H_1', which validates the alternative hypothesis H_1.

4.4. Experiment results – Research question 2

After running all phases of the SoMEN-DC, we attempted to refute the H_2'. As planned, we performed two analyses: a descriptive and qualitative analysis of the two most voted candidates with regard to polls, predictions and the final vote share; and the measurement of the prediction errors by considering the polls as imprecise ground truth. Thus, competitive predictions were those that, when compared with the polls, presented errors below the historical MAE of 2.7 percentage points, within a deviation of 3.00, which is considered the error margin of most polls.

To illustrate the predictions, Figs. 4-7 present the daily predictions, obtained by the MLP-BP PCA model, of the two most voted candidates, as two lines. They also present the polled data as dots and the final election result as the last squares.

The Argentinian data, presented in Fig. 4, demonstrates that for the most voted candidate, Fernandez, after the start of the campaign, the polls overestimated his vote share, and our predictions underestimated, while achieving the exact result on the final prediction day. Moreover, both polls and predictions similarly underestimated the second most voted candidate.

In the Brazilian scenario, presented in Fig. 5, it may be observed that before the last month of the campaign, predictions varied between higher and lower values than the polls, suggesting unbiased values. However, in the last month, almost all the polls for both candidates, and especially for Haddad, were higher than our predictions, and higher than their final vote share. However, our final predictions were extremely close to the final vote share, as presented in Table 7. Although not conclusive, this data suggests that polls may present a bias towards both candidates and overestimate their vote share, which did not occur

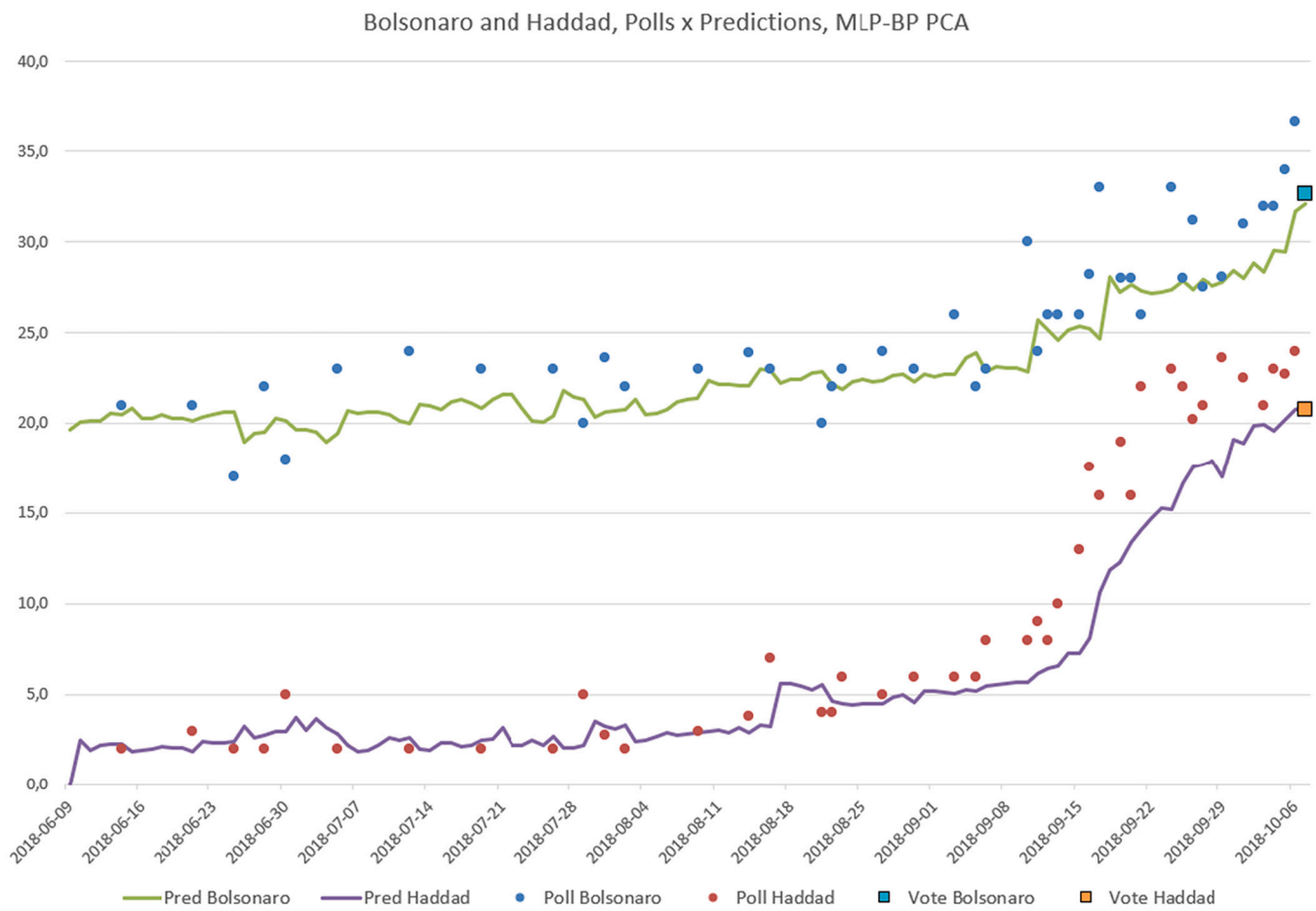

Fig. 5. Polls and predictions in Brazil: Bolsonaro and Haddad, predictions using the MLP-BP PCA model and the final vote share.

with our predictions.

In the Colombian scenario, shown in Fig. 6, both the polls and predictions presented close results nearer to election day, especially for the second most voted candidate. However, many strong outliers were found on polls for both candidates, some of which have been circled in the figure. Furthermore, some spikes were found on predictions for Márques on specific days, which may be explained by the instability of polls on days close to the spikes. However, this effect needs to be investigated further.

Fig. 7 presents the polls and predictions for the Mexican scenario. The polls and predictions for the second most voted candidate, Cortés, remained close during the whole period, with very few exceptions. The high variation in the poll results closer to the elections, from 34.0 to 62.5 percentage points during the last week, made it difficult to undertake accurate predictions for Obrador. Thus, although the predictions obtained results that were competitive with the polls, this result suggests the need for a better approach for poll pruning.

By summarizing this analysis on all countries, we have verified that:

- In Argentina: for the first candidate, results suggest that the prediction bias was lower than the poll bias, since the predictions in the last week were closer than the polls. For the second most voted candidate, predictions and polls seemed equally biased;
- In Brazil, results for both candidates suggest a higher poll bias than prediction bias, since the prediction curve smoothly achieved very accurate final predictions;
- In Colombia, polls and predictions presented similar results, despite some high poll outliers and some prediction spikes, which should be investigated further;
- In Mexico, the polls and predictions were close with regard to the second most voted candidate. However, the high variance in the polls, particularly when close to the elections, and the high errors obtained both by polls and prediction, prevent any conclusions regarding the results.

The MAE comparing all predictions and polls during the campaign is presented in Table 16. The calculation considers polls as the baseline, and the error is the difference between the predicted results (on the days with polls) and polls.

As expected, the linear regression model presented the poorest results, even though the linear regression with PCA presented unexpectedly good results. Moreover, results with neural networks in Argentina, Brazil and Colombia presented a MAE lower than or equal to the usual poll error margin of 3 percentage points, and most values were lower than the historical threshold of 2.7 percentage points, with remarkably low values in Brazil. This data reveals that, on average, these intermediary predictions may be seen as correct predictions of polls. The

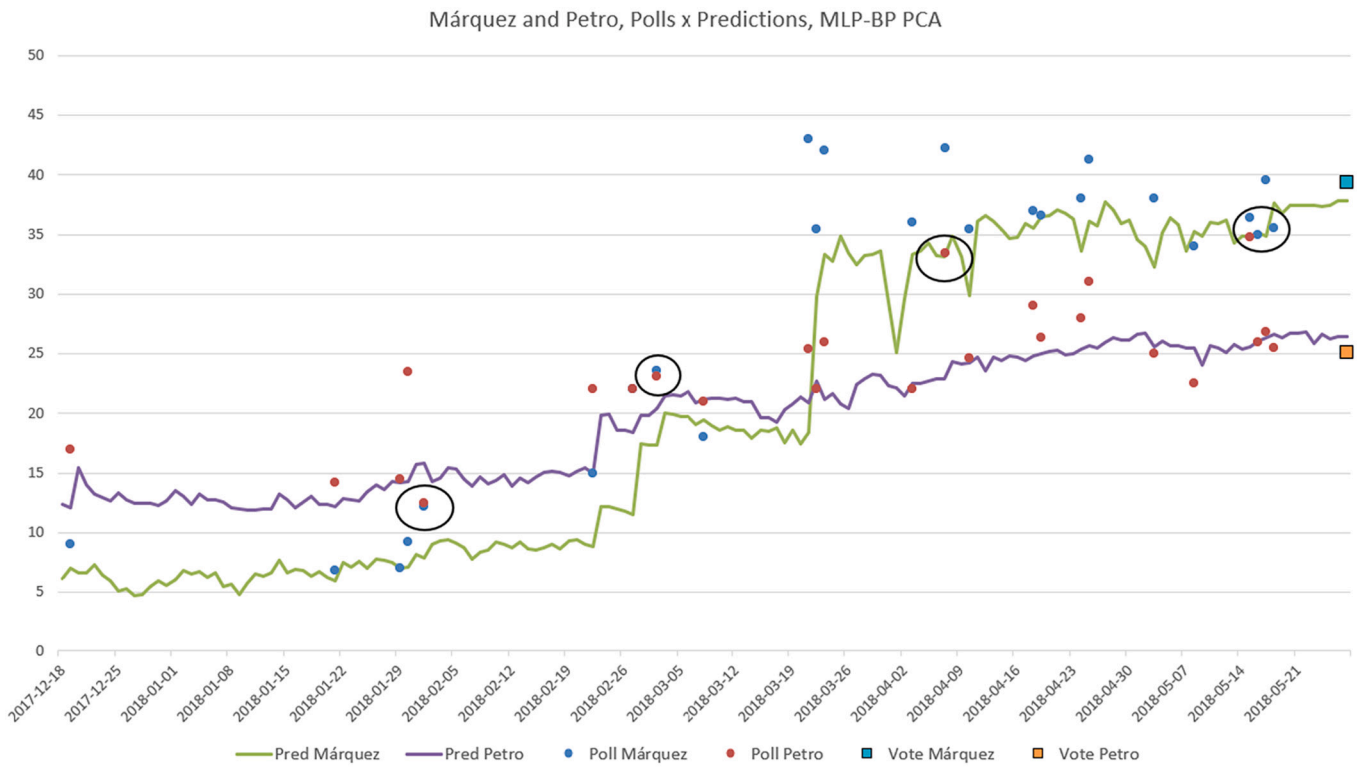

Fig. 6. Polls and predictions in Colombia: Márquez and Petro, predictions using the MLP-BP PCA model and the final vote share.

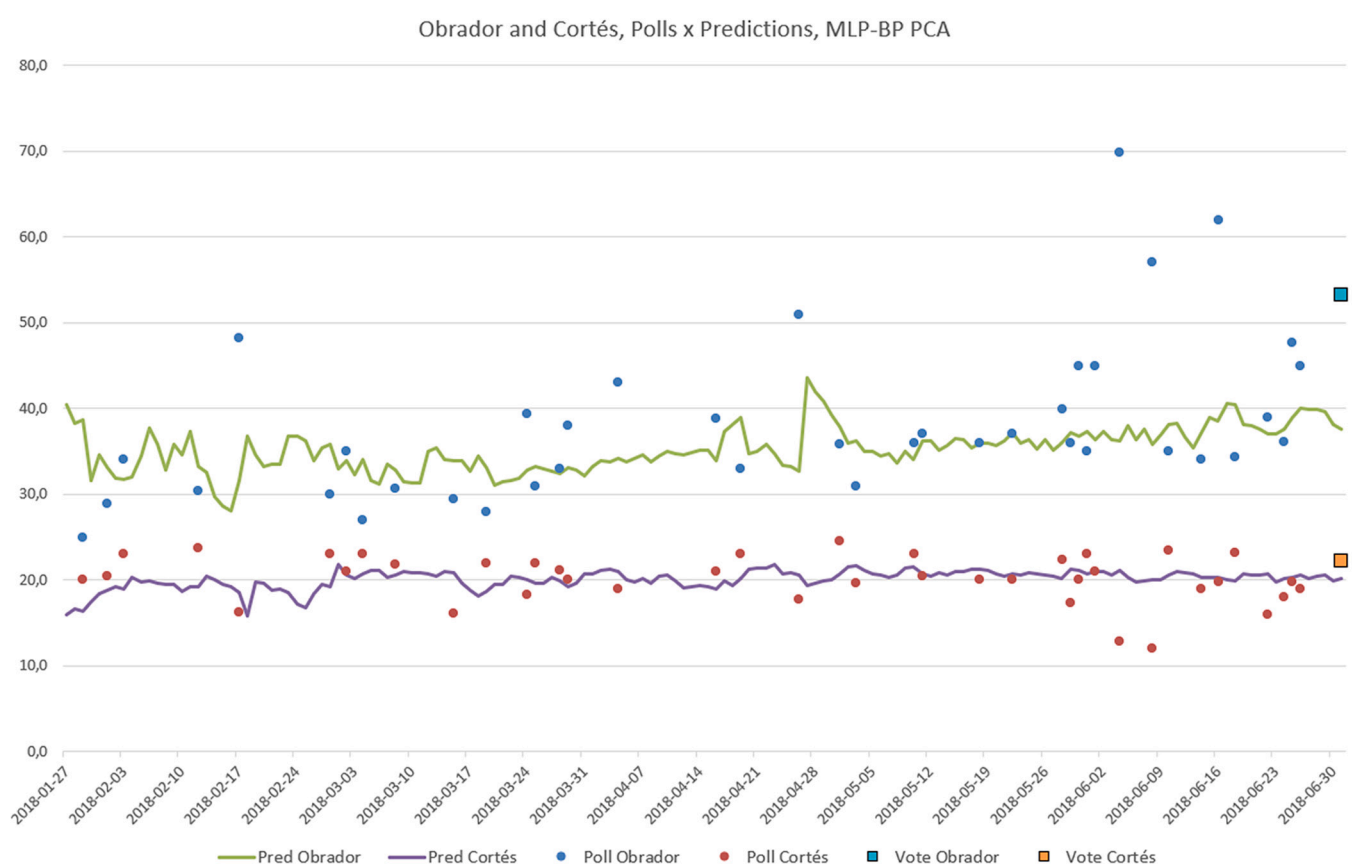

Fig. 7. Polls and predictions in Mexico: Obrador and Cortés, predictions using the MLP-BP PCA model and the final vote share.

Table 16
MAE comparing predictions and polls. This shows how close the predictions are to the polls.

Model	Argentina	Brazil	Colombia	Mexico	Average
LinearRegression	151.93	13.30	46.11	8.70	55.01
LinearRegression PCA	2.60	1.44	3.26	4.01	2.83
MLP-BP	2.21	1.60	2.96	3.56	2.58
MLP-BP PCA	2.29	1.54	2.88	3.59	2.58
GRNN	2.35	1.33	3.00	3.62	2.58
GRNN PCA	2.34	1.35	2.96	3.61	2.57

presented data therefore **rejects** the null hypothesis H_2', which validates the alternative hypothesis H_2.

5. Discussion

This study is not the first claiming to be able to predict elections results. However, the defined process and the presented modeling has addressed many of the challenges found in previous studies, presented in Table 1. The manner in which this proposal has addressed each challenge is discussed below.

The **process challenges** were all addressed by the definition of the SoMEN and the SoMEN-DC processes. The processes were run in four different countries and performed well in different contexts. The SoMEN-DC was defined exactly to enable predictions during a campaign.

The **sampling challenges** were also addressed. The challenge of using social networks as a population sample is addressed by using traditional polls to train the models. The representativity problem of Twitter, as being the sample of all platforms, was addressed by the capability of the proposal to use data from different SM platforms, those that the candidates and the population use most, and is also naturally ready for use even with platforms that are not yet popular.

Moreover, due to a change in the concept of SM performance, from measuring how many people are talking about a candidate, to measuring how many people are paying attention to a candidate and are interacting with the content, the defined approach requires the collection of much less data, a thousand posts from less than a dozen candidates, instead of millions of posts from the entire population. Furthermore, no keyword for data collection needs to be defined. Data collection may also occur for a long period before elections (10 months), and the use of many different window sizes exempts the need of arbitrary decisions regarding how many days of SM data should be collected and used. Lastly, the small amount of data to be collected and processed thereby requires a very low computational power.

In the case of the **modeling challenges,** the high susceptibility to volume manipulation was addressed by training and predicting the candidate results individually. Thus, the model was trained with the specific behavior of the candidate's supporters, and even the behavior of his/her network of BOTs or paid propaganda, should this exist.

Also, to the best of our knowledge, this is the first approach that is capable of crossing data from multiple platforms, particularly Facebook, Instagram, and Twitter.

We also considered the state-of-the-art machine learning and technical modeling. The chosen models, the MLP-BP and the GRNN, although they are not the most recent proposals, are well suited for this context and are capable of presenting nonlinear relationships between inputs and outputs, unlike other approaches based on linear relationships. In particular, the averaged MLP-BP presented remarkable results on small samples in a very recent comparative study (Fernández-Delgado et al., 2019).

In relation to one of the most relevant technical modeling weaknesses, the choice of parameters for the ML models, this study has clearly presented the strategy for parameter choice. The GRNN only needs one parameter, which was found by a grid search strategy. In addition, the MLP-BP parameters were not only chosen based on the characteristics of the sample, but also based on a preliminary study, which compared the chosen parameters with a grid search strategy for the choice of the best parameters and achieved similar results with both strategies.

The challenges of **performance evaluation and scientific rigor** were also addressed. The results were analyzed using the most used metric of the polls domain, and the predictions were compared with the final electoral results and the polls' results. Moreover, historical baselines were used and statistical analyses of the results were proposed and performed.

Although the experiments were carried out in Latin American countries, the approach is sufficiently general to be employed in any country with the freedom to use SM and follow a democratic voting schema. Furthermore, it may be employed in any particular country with data from any popular newsfeed-based social media platform, such as TikTok or even platforms that are not yet popular. Thus, the results may be compared in order to verify whether the same results obtained in this study are achieved under these different conditions.

5.1. Comparison with related work

No studies were found that predicted election results for the same countries during the same years in order to conduct a direct comparison. Therefore, we compared our approach and results with studies containing similar characteristics: predicting elections in the studied countries, using polls for training, and/or predicting elections in more than one country.

Cerón-Guzmán and León-Guzmán (Ceron-Guzman & Leon-Guzman, 2016) studied the 2014 Colombian presidential election using a slightly similar approach. They used a Twitter volume/sentiment approach combined with polls for training linear regression models (Ordinary Least Square, Ridge, Lasso, and Support Vector Regression). However, their results wrongly presented the list of the most voted candidates, and they concluded that "the obtained results show that inference methods based on Twitter data are not consistent…". In terms of the Brazilian elections, (Justino Garcia Praciano et al., 2019) developed an approach also based on volume/sentiment to detect the winner of the second round of the 2014 presidential elections. The authors considered the study as being successful by correctly indicating the winners in 19 out of 26 states. However, since they did not present the vote share, the approach would be better suited to scenarios such as in the U.S.

With regard to studies analyzing multiple elections, Gaurav et al. (Gaurav, Srivastava, Kumar, & Miller, 2013) used a Twitter based volume approach to predict elections in Ecuador, Paraguay, and Venezuela in 2013. They concluded that "counting the tweets that mention a candidate's conventional name is not sufficient to obtain good predictions." Anjaria et al. (Anjaria & Guddeti, 2014) employed four ML techniques (SVM, Naïve Bayes, maximum entropy, and MLP-BP) for a Twitter sentiment analysis to predict the 2012 U.S. presidential elections and the 2013 Karnataka (India) state assembly elections. Their best result in the U.S. presented a MAE of 3.44, but a MAE of 13.60 in the Indian elections largely differed from the actual vote share.

The study we consider to be the closest to ours was presented by Tsakalidis et al. (Tsakalidis, Papadopoulos, Cristea, & Kompatsiaris, 2015). They used Twitter data to predict the 2014 EU election results in Germany, the Netherlands, and Greece. Despite using the traditional volume and sentiment model, they used 11 volume/sentiment derived variables combined with one poll-based feature for training three regressors (linear regression, Gaussian process, and sequential minimal optimization), and the output average of the three was used as the final estimate in a combined regressor. Their approach achieved a good performance and obtained an MAE below 2.0 in the three countries, although they did not compare the results with polls. Notwithstanding the good results, the study presented well-known drawbacks: the

inclusion/exclusion criteria for keywords selection and the time window selection are quite arbitrary, and they recognized that no more than 1 percent of the public tweets had been gathered. Even with this limitation, their approach collected an average of 350,175 posts per country. Nevertheless, the approach presented in this paper does not present these problems of arbitrary choices, is able to collect all the required data, and needs to gather only a fraction of the posts (16,851 posts by country, 4.8% of data).

6. Concluding remarks

This study has investigated a new approach of using SM data for predicting elections. We adopted a new way of measuring performance on SM. Instead of considering how many people are talking about a candidate, the new approach considers how many people are paying attention to a candidate. We have also presented a new process, which is capable of being repeated in different elections with minor adjustments. A ML model was also created composed of an ensemble of artificial neural networks. Moreover, the framework (process and model) is able to make predictions with data from different SM platforms, even those that have not as yet become popular, using nonlinear ML models.

The proposed framework has attempted to deal with the main challenges recognized in previous studies, including the challenges presented by process, sampling, modeling, and performance evaluation and scientific rigor.

Experiments were performed with data from recent presidential elections in four of the most populous Latin American countries, Argentina, Brazil, Colombia, and Mexico. The results show that it was possible to use the framework to achieve a high level of accuracy in predicting the final vote share of the candidates, with competitive or better results than traditional polls, as well as to achieve results closer to polls during the campaign, in a daily prediction setting.

We highlight that we made public two predictions before the final release of results by applying early versions of this approach: the prediction of the 2019 Argentinian election was published on Facebook[7] on election day and attained better results than the considered polls at that time, and the prediction of the 2020 U.S. elections was published in LinkedIn[8] on the day of elections, obtaining an MAE error lower than the RCP[9] poll average. These results reinforce the claims that the proposed approach may be used in predicting elections in real-time, rather than only working with data from the past.

This was a multidisciplinary research effort and involved knowledge from the areas of polling and electoral predictions, social media studies, and machine learning. Thus, we emphasize the specific contributions to these areas.

For the subject of **electoral predictions**, the central feature of many governments and democracies, we proposed an approach that was able to complement traditional polling. As demonstrated in this study, it may be used to perform nowcasting on days with no polling, or may be included in the traditional methodology used by pollsters in order to update their methods to use data from both interviews and SM. It also enables candidates and political parties to obtain an estimate of their performance just by using their SM data and previous polls, and may be used as a quick, cheap approach to obtaining daily predictions.

For the **machine learning** area, this work has studied and applied ML for a new, as yet underexplored context. This context presents very specific challenges, such as the lack of historical data, the rapid change of context, a small number of samples and the existence of only one actual labeled data, the final result, which is the aim of prediction. Also, this context presents external interferences, such as the existence of bots or even the natural difference of the behavior of supporters on SM. We consider the proposals, including the set of features, the training of each candidate individually, and the ensemble of many ANN for different windows, as contributions to the area of ML.

For **social media** studies, this work has studied the possibility of finding nonlinear correlations between online behavior and offline outcomes. This kind of result is difficult to achieve by social scientists because they frequently face difficulties in accessing and processing the high volume of data needed for these calculations. Moreover, additional studies may help to derive how the candidates from these countries used SM as a campaign platform, how users were engaged in SM, and how these countries were polarized during that election.

For future research, we highlight the possibility of applying the proposed framework to other electoral contexts, such as presidential elections in other countries and other types of elections. We also consider that neither the process nor each of its phases are final and therefore, may be improved upon or fine-tuned. The model may also be adjusted with different options, such as different approaches to dimensionality reduction or feature selection, as well to define more effective ways of pruning polls and excluding outliers, all of which may lead to an improvement in the results. Different ML strategies to improve the accuracy of predictions, such as online learning (Losing et al., 2018), recurrent neural networks (Medsker & Jain, 2001), few-shot learning (Wang, Yao, Kwok, & Lionel, 2020), and the use of concept drift (Lu et al., 2018), may also be promising, as well as the use of explainable AI (Goebel et al., 2018) for better interpretability of results.

CRediT authorship contribution statement

Kellyton Brito: Conceptualization, Methodology, Software, Formal analysis, Investigation, Writing – original draft. **Paulo Jorge Leitão Adeodato:** Conceptualization, Methodology, Validation, Writing – review & editing, Supervision.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A

[7] Available at https://www.facebook.com/notes/404228243926826/. Viewed on: January 13, 2022.

[8] Available at https://www.linkedin.com/pulse/prevendo-o-resultado-das-elei%C3%A7%C3%B5es-de-hoje-nos-estados-kellyton-brito/. Viewed on: February 08, 2022.

[9] https://www.realclearpolitics.com/epolls/2020/president/us/general_election_trump_vs_biden-6247.html. Viewed on: January 13, 2021.

Metadata			Features: 7-day window																		Target
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Jair Bolsonaro	7	Datafolha	31/01/2018	26	1240519	140364	313436	47712	5399	12055	25	118256	35344	4730	1414	12	537259	15602	44772	1300	1800
Jair Bolsonaro	7	MDA	04/03/2018	23	537136	31882	147563	23354	1386	6416	26	80495	19540	3096	752	10	396580	8838	39658	884	2000
Jair Bolsonaro	7	MDA	13/03/2018	17	551722	124801	143647	32454	7341	8450	14	57436	12739	4103	910	8	259977	8198	32497	1025	1830
Jair Bolsonaro	7	Datafolha	14/04/2018	18	623261	36655	198629	34626	2036	11035	22	141874	40562	6449	1844	12	619894	16871	51658	1406	1700
Jair Bolsonaro	7	DataPoder360	20/04/2018	31	1191262	95279	450268	38428	3074	14525	23	136287	38895	5926	1691	17	572379	12814	33669	754	2000
Jair Bolsonaro	7	Paraná Pesquisas	03/05/2018	24	630387	69941	170844	26266	2914	7119	23	120275	25267	5229	1099	10	373893	8185	37389	819	2050
Jair Bolsonaro	7	Ipespe	24/05/2018	23	751025	54831	292605	32653	2384	12722	22	111117	24830	5051	1129	12	607902	12014	50659	1001	2400
Jair Bolsonaro	7	DataPoder360	01/06/2018	23	985157	63514	540122	42833	2761	23484	25	223428	56971	8937	2279	16	847068	17931	52942	1121	2100
Jair Bolsonaro	7	Ipespe	07/06/2018	29	548229	44408	154040	18904	1531	5312	18	84539	18959	4697	1053	13	542434	11354	41726	873	2200
Jair Bolsonaro	7	Datafolha	08/06/2018	29	558771	56770	164586	19268	1958	5675	22	109107	24140	4959	1097	15	622199	12357	41480	824	1900
	...							...					...					...			...
Jair Bolsonaro	7	Paraná Pesquisas	05/10/2018	56	7143656	2436552	2238079	127565	43510	39966	123	2199156	644303	17879	5238	38	15378951	565259	404709	14875	3490
Jair Bolsonaro	7	MDA	06/10/2018	58	7444892	2741148	2352175	128360	47261	40555	120	2098763	611363	17490	5095	39	16263326	543275	417008	13930	3670
Jair Bolsonaro	***7***	***ActualVoteShare***	***07/10/2018***	***59***	***7581827***	***3251916***	***2435191***	***128506***	***55117***	***41274***	***122***	***2251065***	***629977***	***18451***	***5164***	***39***	***16120375***	***516437***	***413343***	***13242***	***3263***

Metadata			Features: 28-day window																		Target
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Jair Bolsonaro	28	DataPoder360	01/06/2018	98	2864356	252646	1151735	29228	2578	11752	88	527583	127583	5995	1450	53	2481077	51218	46813	966	2100
Jair Bolsonaro	28	Ipespe	07/06/2018	109	3022926	271624	1199495	27733	2492	11005	89	536517	129579	6028	1456	55	2630693	53418	47831	971	2200
Jair Bolsonaro	28	Datafolha	08/06/2018	109	3000671	267305	1203022	27529	2452	11037	92	555502	133876	6038	1455	57	2700851	53549	47383	939	1900
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Jair Bolsonaro	28	Paraná Pesquisas	05/10/2018	139	15459647	3693880	4473254	111220	26575	32182	248	5544029	1376612	22355	5551	102	36133969	1348619	354255	13222	3490
Jair Bolsonaro	28	MDA	06/10/2018	145	15946010	4017560	4724670	109972	27707	32584	251	5534535	1384541	22050	5516	107	38796291	1440976	362582	13467	3670
Jair Bolsonaro	***28***	***ActualVoteShare***	***07/10/2018***	***151***	***16780732***	***4586624***	***5030580***	***111131***	***30375***	***33315***	***274***	***5914169***	***1503145***	***21585***	***5486***	***111***	***40577829***	***1483609***	***365566***	***13366***	***3263***

Fig. A.1. An example of two generated datasets, with 7 and 28 windows.
*The first 4 columns are the metadata, not used for training/test. The last column, "Share", is the target, the vote share obtained by the candidate on that poll. It is multiplied by 100 to facilitate calculations. Thus, a vote share presented as 1800 is 18.00. The other columns are the features. The last line is the actual vote share of the the candidate on the elections.

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Kellyton Brito is an Adjunct Professor in the Departamento de Computação of Universidade Federal Rural de Pernambuco, Brazil. He holds an M.Sc degree in Computer Science with a focus on Software Engineering and a PhD degree in Computer Science with a focus on Computational Intelligence, both from the Universidade Federal de Pernambuco, Brazil. He is a researcher at InnovaGovLab, and his research is focused on studying and applying ICT in a collaboration between public and private organizations, including subjects such as machine learning, elections forecasting, social media, open government data, and software design and architecture. More information: https://kellyton.com.br/

Paulo Jorge Leitão Adeodato holds a tenure-track position as a professor in Centro de Informática of Universidade Federal de Pernambuco, Brazil, and a PhD in mathematics from King's College London (1997). In 2017, he was a Visiting Scholar at Stanford Graduate School of Education, USA, and a Visiting Professor at Stanford School of Medicine, USA. His research interest includes the development of decision support systems with explainable artificial intelligence, quality assessment of binary classifiers, and startup entrepreneurship and investment. Prof. Adeodato's team awards include the first-place Winner of the FINEP Innovation Award for Small Entrepreneurship in the Brazilian Northeast (2013, Brazilian Innovation Agency), 1st Place in the Time Series Forecasting Competition (IEEE + Kaggle, 2012, Madrid, Spain), the second place in the NN3 Time Series Forecasting Competition (IEEE - IJCNN 2007, Orlando, USA), and the second place in the PAKDD Data Mining Competition (2007, Nanjing, China).